

Day 7 Student Handout

Geometry Summer Credit | Quadrilaterals and Coordinate Tests

Name:	Date:	Period:
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Learning Targets

I can use properties of parallelograms, rectangles, rhombi, squares, and trapezoids. I can classify quadrilaterals on the coordinate plane using slope, distance, and midpoint.

Lesson 7A: Parallelograms and Special Quadrilaterals

Warm-Up

1.	Find the slope between A(1, 3) and B(7, 6).
2.	Find the distance between C(-2, 1) and D(4, 9).
3.	Find x if opposite sides of a parallelogram are $2x + 5$ and 19.

Concept Launch

Quadrilateral Family

A parallelogram has two pairs of parallel opposite sides. Rectangles, rhombi, and squares are special parallelograms. A trapezoid has one pair of parallel sides.

Problem	Work / notes
Parallelogram ABCD has $AB = 14$ and $BC = 9$. Find CD and AD.	Opposite sides: Answers:
Parallelogram ABCD has angle A = 68 degrees. Find angle C and angle B.	Opposite angles: Consecutive angles:
A rectangle has diagonals $AC = 3x + 4$ and $BD = 25$. Find x.	Rectangle diagonals: Equation: Answer:

A rhombus has one side length 11. What are the other side lengths?	Rhombus sides: Answers:
A square has side length 8. What must be true about its sides and angles?	Sides: Angles:

Property Summary

Most Useful Facts

Parallelogram: opposite sides parallel and congruent; opposite angles congruent; consecutive angles supplementary; diagonals bisect each other. Rectangle: parallelogram with four right angles and congruent diagonals. Rhombus: parallelogram with four congruent sides. Square: both rectangle and rhombus.

4.	Find x if consecutive angles of a parallelogram are $5x$ and $3x + 20$.
5.	A rectangle has diagonal lengths $4x - 3$ and $2x + 9$. Find x .

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Lesson 7B | Coordinate Tests for Quadrilaterals

Lesson 7B: Coordinate Evidence

How to Classify with Coordinates

Use slope to test for parallel or perpendicular sides. Use distance to test for congruent sides or diagonals. Use midpoint to test whether diagonals bisect each other.

Problem	Work / notes
A(0, 0), B(6, 0), C(8, 4), D(2, 4). Use slopes to decide if ABCD is a parallelogram.	Slope AB: Slope CD: Slope BC: Slope AD: Conclusion:
P(0, 0), Q(4, 0), R(4, 3), S(0, 3). Classify the quadrilateral.	Slopes: Side lengths: Conclusion:
M(-2, 1), N(2, 4), O(5, 0), P(1, -3). Use distance to test whether all sides are congruent.	MN: NO: OP: PM: Conclusion:
A parallelogram has diagonals with midpoints (3, 2) and (3, 2). What does that evidence show?	Diagonals: Conclusion:

Practice

6.	A quadrilateral has side slopes 0, undefined, 0, undefined and all angles right. What type could it be?
7.	A quadrilateral has two pairs of parallel sides and all four sides congruent. What type is it at least?
8.	Find the midpoint of diagonal AC if A(-1, 5) and C(7, -3).

9.	Explain why slope is useful for classifying quadrilaterals.
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